

Summary

Computer Science student with hands-on experience building **full-stack apps** across **TypeScript/Next.js** and **Java/Spring Boot**, including **RESTful APIs**, **relational databases**, **authentication**, and **cloud deployment**. Experienced in designing and implementing **layered architectures** and developing reliable software through **TDD** and **automated testing**. Built production-deployed applications using **PostgreSQL**, **Docker**, **AWS**, and **Vercel**. Strong collaborator with a team-first mindset and precise attention to logic for correct implementation

Education

Honours Bachelor of Science in Computer Science, York University, Toronto

Expected Oct 2027

Relevant Projects

Hierarchical Document Workspace App | [Site](#)

- Built and deployed a **full-stack hierarchical document workspace app** via **Next.js**, **React**, **TypeScript** & **Tailwind** with **authentication**, **block-based rich-text editing**, **nested page structure** & **full-text search**
- Designed a **layered architecture** separating **Server Actions**, **service-layer business logic**, **Prisma ORM**, and **React server/client components** for **maintainability** via **separation of concerns**
- Implemented **session-based authentication** (BetterAuth) and **ownership-based authorization** across page operations to enforce **secure, private data access**
- Developed **full-text search** via **GIN indexing**, **pagination**, **debounced search**, **highlighted matches**, **relevance ranking** & **stored vectors** to **avoid recomputation** for **query performance**
- Integrated **BlockNote** with **custom blocks and menus**, **JSON document persistence**, **breadcrumb page navigation**, **debounced autosave** to reduce unnecessary **database writes**
- Designed **relational schema** & integrated **PostgreSQL** via **Prisma** for persistent storage, deploying the **production application** on **Vercel**

Bank Mobile App

- Built and deployed a **RESTful Java full-stack banking system** via **Spring Boot** & **PostgreSQL**, using a **layered architecture** (controller, service, repository) for **maintainability**
- Designed **relational schema** & integrated **PostgreSQL** via **Hibernate ORM** for **persistent storage**
- Built **client UI** in **Android Studio** & integrated **5 REST endpoints** by **analyzing user interface flow**, enabling core banking operations (account creation, deposit, withdraw, transfer, statement generation)
- Applied **TDD**, **JUnit**, **Mockito**, and **MockMvc**, covering **15 automated unit, controller** and **service integration-level tests** for **code reliability** & **efficient debugging** through **early defect detection**
- Integrated **business workflow validation** in a **core service layer** before delegating operations, **preventing invalid operations** from propagating deeper into the system
- **Dockerized Spring Boot API** & deployed to **AWS Lightsail**, for **mobile app access to REST APIs**

Driving Simulation

- Built a **Python-based** driving simulation with **OOP** to model user progression on a virtual road
- Utilized **PyGame API** to develop application logic & UI elements (e.g. vehicles, dynamic progress bar)
- Designed **dynamic generation of AI vehicles** & **rule-based movement logic** with **attention to detail** to produce intentional collision scenarios, improving **user engagement**
- Developed **3 termination conditions** to handle **state transitions** involving **bounding box collision detection** with user & AI, screen boundary violations, and finish-line completion logic

Technical Skills

Languages: Java, Python, Typescript, SQL, HTML, CSS, Bash, C

Backend: Spring Boot, Node.js, Hibernate (Spring Data JPA), Prisma, BetterAuth, JUnit, MockMvc, Mockito

Frontend: Next.js, React, Tailwind, Bootstrap, Figma

DevOps/Cloud: Docker, AWS Lightsail, Vercel, Azure

Databases & Other: PostgreSQL, SQL Server, Git, GitHub, Android Studio

Volunteering

Orientation Logistics Volunteer

August 2019

University of Toronto

- Assembled **8000+ student orientation kits** with student supplies in a **fast-paced, deadline-driven environment** through **constant communication** and **teamwork** with **100+ volunteers**
- Collaborated with **volunteers** to manage **5000+ by-products** from packing to maintain an **organized workspace**